of the Human Genome Project. The mission of the ELSI program was to identify and address issues raised by genomic research that would affect individuals, families, and society. A percentage of the Human Genome Project budget at the National Institutes of Health and the U.S. Department of Energy was devoted to ELSI research.

Genetic Engineering Appraisal Committee (GEAC)

GEAC is an apex body under Ministry of Environment, Forests and Climate change for regulating manufacturing, use, import, export and storage of hazardous microbes or genetically modified organisms (GMOs) and cells in the country. It was established as an apex body to accord approval of activities involving large scale use of hazardous microorganisms and recombinants in research and industrial production. The GEAC is also responsible for approval of proposals relating to release of genetically engineered organisms and products into the environment including experimental field trials.

5.7 Future of Biotechnology

Biotechnology has become a comprehensive scientific venture from the point of academic and commercial angles, within a short time with the sequencing of human genome and genome of some important organisms. The future developments in biotechnology will be exciting. Thus the development in biotechnology will lead to a new scientific revolution that would change the lives and future of people. Like industrial and computer revolution, biotechnological revolution will also promise major changes in many aspects of modern life.

Summary

Tissue culture is the in vitro asceptic culture of cells, tissues or organs into whole plants under controlled nutritional and environmental conditions. A German physiologist Gotllieb Haberlant in 1902 for the first time attempted to culture plant cells in artificial medium, hence he was regarded as father of Tissue culture. Tissue culture mainly based on the concepts totipotency, differentiation, redifferentiation and dedifferentiation. Plant tissue culture technique involves selection of explants, sterilization, media preparation, maintaining culture condition, callus formation, embryogenesis or organgenesis and hardening. Based on the explants chosen the types of tissue culture are organ culture, meristem culture, protoplast culture and cell suspension culture. From the explants, plants can be regenerated by somatic embryogenesis or organgenesis is said to be plant regeneration pathway. Some of the main applications of tissue culture are production of somatic hybrids, artificial seeds, disease resistant and stress resistant plants, germplasm conservation, micropropagation and production of secondary metabolites. Intellectual Property Right (IPR) is primarily aimed at patents, copyrights, trade secret and trademark given to the discoverer / inventor for the commercial production of transformed micro organisms or plants. Biosafety is the prevention mechanism to protect harmful incidents due to biohazards or pathogens. Bioethics dealt with ethical issue emerging from biotechnological advancement. ELSI program addresses issues related to genenomic research. GEAC (Genetic Engineering Appraisal Committee) is a regulatory authority for release of genetically modified products or organisms into the environment.

Evaluation

Choose the correct answer from the given option:

1. Totipotency refers to
   a) capacity to generate genetically identical plants.
   b) capacity to generate a whole plant from any plant cell / explant.
   c) capacity to generate hybrid protoplasts.
   d) recovery of healthy plants from diseased plants.

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